

House Price Prediction Using Machine Learning

Internship Project Report

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Project Title: House Price Prediction

Technology: Python, Machine Learning, Linear Regression

1. Introduction

House Price Prediction is a machine learning project used to estimate house prices based on different features such as crime rate, number of rooms, taxes, and location-related information. This project uses Linear Regression to predict housing prices accurately.

2. Objective

The main objective of this project is to build a predictive model using historical housing data and estimate house prices using machine learning techniques.

3. Dataset Description

The project uses the Boston Housing Dataset. The dataset contains multiple numerical features including crime rate, average number of rooms, tax rate, pupil-teacher ratio, and median house value.

4. Technologies Used

Python Programming Language

Pandas

NumPy

Matplotlib

Scikit-learn

Jupyter Notebook / VS Code

5. Methodology

1. Load Dataset
2. Data Cleaning
3. Feature Selection
4. Train-Test Split
5. Model Training using Linear Regression
6. Model Evaluation using RMSE and R^2 Score
7. Prediction and Model Saving

6. Linear Regression Formula

$$y = \beta_0 + \beta_1x_1 + \beta_2x_2 + \dots + \beta_nx_n$$

7. Python Code

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

data = pd.read_csv("housing.csv")

X = data.drop("medv", axis=1)
y = data["medv"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

print("R2 Score:", r2_score(y_test, y_pred))
```

8. Output

The model successfully predicts house prices using Linear Regression. The evaluation metrics such as RMSE and R² Score indicate good prediction performance.

9. Conclusion

This project demonstrates how machine learning can be used in real-world applications like property price prediction. The model provides accurate results and helps understand regression analysis.

10. Future Scope

The project can be improved using advanced machine learning algorithms such as Random Forest, XGBoost, or Deep Learning for better prediction accuracy.